

ODE System: Gut Immune-Microbe Model

Helper Functions

Logistic threshold (sigmoid gate):

$$f_{\text{th}}(x; \theta, c) = \begin{cases} \frac{1}{1 + e^{-c(x-\theta)}} & x > \theta \\ 0 & x \leq \theta \end{cases} \quad (1)$$

AMP killing rate:

$$f_A(E) = \tau_A \frac{E}{E_{\max}} \left(1 - \frac{E}{E_{\max}} \right) \quad (2)$$

Barrier breach rate:

$$f_B(E; \gamma_B) = \gamma_B E \quad (3)$$

ROS-mediated microbial killing:

$$f_R(R) = f_{\text{th}}(R; \theta_M, c_M) \quad (4)$$

Engulfment by macrophages:

$$f_E(M_i) = \alpha_i M_i \quad (5)$$

M1 activation signal:

$$f_1(D) = f_{\text{th}}(D; \theta_1, c_1) \quad (6)$$

State Equations

Lumen

Pathogens (lumen):

$$\frac{dP^L}{dt} = r_P^L P^L \left(1 - \frac{P^L}{K_P^L} \right) - P^L f_A(E) - P^L f_B(E; \gamma_{B,P}) \quad (7)$$

Commensals (lumen):

$$\frac{dC^L}{dt} = 0 \quad (\text{placeholder}) \quad (8)$$

Lamina Propria

Pathogens (LP):

$$\frac{dP^{LP}}{dt} = r_P^{LP} P^{LP} \left(1 - \frac{P^{LP}}{K_P^{LP}} \right) - P^{LP} f_R(R) r_{\text{ros}} - P^{LP} (f_E(M_0) + f_E(M_1) + f_E(M_2)) + (P^L - P^{LP}) f_B(E; \gamma_{B,P}) \quad (9)$$

Commensals (LP):

$$\frac{dC^{LP}}{dt} = r_C^{LP} C^{LP} \left(1 - \frac{C^{LP}}{K_C^{LP}} \right) - C^{LP} f_R(R) r_{\text{ros}} - C^{LP} (f_E(M_0) + f_E(M_1) + f_E(M_2)) + (C^L - C^{LP}) f_B(E; \gamma_{B,C}) \quad (10)$$

Epithelial Damage

$$\frac{dE}{dt} = \tau_{P,A} P^L (E_{\text{max}} - E) + \tau_{P,B} P^{LP} (E_{\text{max}} - E) + f_{\text{th}}(R; \theta_E, c_E) (E_{\text{max}} - E) - \mathbf{1}_{E>0} \kappa_E \quad (11)$$

Danger Signals

$$\frac{dD}{dt} = \beta_D E + \beta_M (P^{LP} + C^{LP}) + \beta_A P^{LP} - \tau_D D \quad (12)$$

Reactive Oxygen Species

$$\frac{dR}{dt} = \beta_1 M_1 - \tau_R R \quad (13)$$

Macrophages

M0 (naïve):

$$\frac{dM_0}{dt} = r_{\text{rec}} D (M_{\text{max}} - (M_0 + M_1)) - M_0 f_1(D) \quad (14)$$

M1 (pro-inflammatory):

$$\frac{dM_1}{dt} = \tau_{1,a} M_0 f_1(D) - \tau_{1,d} M_1 \quad (15)$$

M2 (anti-inflammatory):

$$\frac{dM_2}{dt} = 0 \quad (\text{placeholder}) \quad (16)$$

Parameters

Symbol	Description	Value
r_P^L	Pathogen growth rate (lumen)	0.1
K_P^L	Pathogen carrying capacity (lumen)	100
K_C^L	Commensal carrying capacity (lumen)	100
r_P^{LP}	Pathogen growth rate (LP)	0.0
r_C^{LP}	Commensal growth rate (LP)	0.0
K_P^{LP}	Pathogen carrying capacity (LP)	100
K_C^{LP}	Commensal carrying capacity (LP)	100
τ_A	AMP killing parameter	0.1
$\gamma_{B,P}$	Barrier breach rate (pathogens)	5×10^{-4}
$\gamma_{B,C}$	Barrier breach rate (commensals)	1×10^{-6}
θ_M	ROS threshold (microbes)	250
c_M	ROS killing Hill coefficient	0.1
r_{ros}	ROS killing rate	0.001
α_0	Engulfment rate by M_0	0
α_1	Engulfment rate by M_1	10^{-2}
α_2	Engulfment rate by M_2	0
$\tau_{P,A}$	Apical damage rate	0.1
$\tau_{P,B}$	Basolateral damage rate	0.001
θ_E	ROS damage threshold (epithelium)	100
c_E	ROS damage Hill coefficient	0.1
κ_E	Epithelial recovery rate	0.1
β_D	DAMP production rate	0.01
β_M	DAMP from microbes in LP	0.01
β_A	PAMP from pathogens in LP	0.01
τ_D	Danger signal decay rate	0.01
β_1	ROS production by M_1	0.1
τ_R	ROS decay rate	0.01
θ_1	M1 activation threshold	1
c_1	M1 activation Hill coefficient	0.1
$\tau_{1,a}$	M1 activation rate	0.1
$\tau_{1,d}$	M1 deactivation rate	0.001
E_{max}	Maximum epithelial damage	100
M_{max}	Maximum macrophage pool	100
r_{rec}	Macrophage recruitment rate	10^{-4}